Roll No.: ....

ODD END SEMESTER EXAMINATION DECEMBER 2025

Name of the Program: Bachelor of Technology (Biotechnology) Semester: III

Name of the Course: Structural Biology Course Code: TBT306b

Time: 3 Hours **Maximum Marks: 100**

Note:

i. **All questions are compulsory.**
ii. **Answer any two sub questions among a, b & c in each main questions**
iii. **Total marks in each main question are twenty.**
iv. **Each questions carries 20 marks**

	(20 Marks)	
Q1		
(a)	Describe the emergence of structural biology in the post-NGS era and discuss its importance in modern biological research.	CO1
(b)	Explain the "Sequence–Structure–Function" relationship in biomolecules with suitable examples.	CO1
(c)	Evaluate how structural biology bridges the gap between genomics and functional biology.	CO1
	(20 Marks)	
Q2		
(a)	Explain in detail the different levels of protein structure with appropriate diagrams.	CO2
(b)	Explain why proteins are considered the primary target molecules for structural studies compared to other biomolecules.	CO2
(c)	Define the term protein crystallization. What are the essential requirements for successful crystal formation? Also explain the basic steps and methods involved in protein crystallization.	CO3
	(20 Marks)	
Q3		
(a)	Describe the role of motifs and domains in determining protein structure and function.	CO3
(b)	What are enzyme-ligand interactions? Discuss how enzymes develop 'magic pockets' for catalysis.	CO4
(c)	Evaluate how mutations at the active site or within structural domains can alter the enzyme's function.	CO3
	(20 Marks)	
Q4		
(a)	Discuss how covalent and non-covalent interactions contribute to the stability of macromolecular structures.	CO3
(b)	Explain the structure and organization of a Protein Data Bank (PDB) file. What type of information can be extracted from it?	CO4
(c)	Propose a stepwise workflow for studying a protein's structure—from data retrieval to visualization and analysis—using open-access bioinformatics tools.	CO4
	(20 Marks)	
Q5		
(a)	Discuss the phase problem in X-ray crystallography. Why is it critical for determining molecular structures?	CO4
(b)	Define Molecular Dynamics (MD) simulation. Explain the need for molecular simulations in understanding biomolecular structure, function, and dynamics.	CO5
(c)	Evaluate and compare commonly used MD simulation software packages such as GROMACS, AMBER, CHARMM, and NAMD, highlighting their key features and applications	CO5